CprE/SE 4910 WEEKLY REPORT 07

10/27/2026 – 11/03/2026

Group number: sdmay27-14

Project title: 3D Plan Visualization: Automated Building Reconstruction and Walkthrough Generator

Client: Buildertrend

Faculty Advisor: Dr. Islam

Team Members/Role:

- ***Nakshatra Gupta - Communication Lead***
- ***Rishiraj Parmar - Organization Lead***
- ***Ryan McNab - Team Meetings Lead***
- ***Daniel Hargrave - Computer Vision Lead***
- ***Jase Beaubien - Graphics Lead***

Weekly Summary

Work was done on all of the parts of the project. The first non-prototype version of the model was completed and the json structure was updated.

Past Week Accomplishments

- Group Work:
    - Started working on the lighting talk that is going to be given next week. Discussed different topics to add and future work for each team member.
- Nakshatra Gupta:
    - Took on ownership of the JSON schema that connects computer vision and graphics, as assigned after the scope discussion with Buildertrend and the TA.
    - Made a corrected schema package (v0.2). It adds an optional schema_version field so any file shows which draft it follows. The Python data model now reads stairs, scale, story elevation and opening heights, which it previously ignored. The validator and demo were tested on the package.,

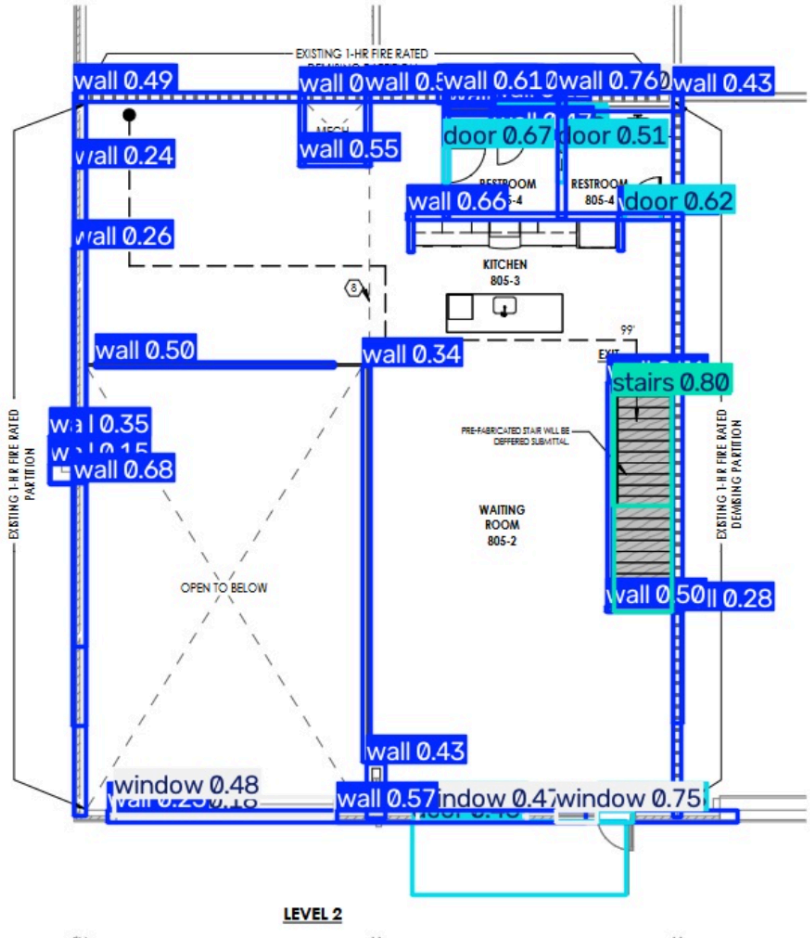


```
"building_id": "sample_001",
"schema_version": "0.2",
"source_file": "sourcefile.pdf",
"scale": {"calibrated": false, "method": "assumed"},
"stories": [
  {
    "story_index": 0,
    "elevation_m": 0.0,
    "base_height_m": 2.7,
    "walls": [
      {"id": "w1", "start": [0.0, 0.0], "end": [6.8, 0.0], "thickness_m": 0.2, "exterior": true},
      {"id": "w2", "start": [6.8, 0.0], "end": [6.8, 11.0], "thickness_m": 0.2, "exterior": true},
      {"id": "w3", "start": [6.8, 11.0], "end": [0.0, 11.0], "thickness_m": 0.2, "exterior": true},
      {"id": "w4", "start": [0.0, 11.0], "end": [0.0, 0.0], "thickness_m": 0.2, "exterior": true},
      {"id": "w5", "start": [3.4, 0.0], "end": [3.4, 11.0], "thickness_m": 0.1, "exterior": false}
    ],
    "rooms": [
      {"id": "r1", "name": "Bedroom 1", "boundary_wall_ids": ["w1", "w5", "w3", "w4"], "area_m2": 18.6},
      {"id": "r2", "name": "Bedroom 2", "boundary_wall_ids": ["w5", "w2", "w3"], "area_m2": 21.9}
    ],
    "openings": [
      {"id": "d1", "type": "door", "wall_id": "w4", "position_m": 5.2, "width_m": 0.9},
      {"id": "win1", "type": "window", "wall_id": "w1", "position_m": 1.5, "width_m": 1.2, "sill_height_m": 0.9, "height_m":
      {"id": "win2", "type": "window", "wall_id": "w2", "position_m": 6.0, "width_m": 1.2, "sill_height_m": 0.9, "height_m":
    ],
    "stairs": [
      {"id": "s1", "outline": [[0.5, 9.0], [2.5, 9.0], [2.5, 10.5], [0.5, 10.5]], "direction": "up", "step_count": 14}
    ]
  }
],
"roof": {
  "type": "hip",
  "pitch_deg": 30
}
```

- Rishiraj Parmar:
    - Worked on calculating floor base elevations from story heights to determine the vertical placement of each floor in the reconstruction prototype.
- Ryan McNab:
    - Looked for more commercially available datasets (ResPlan and Swiss Dwellings) for training use, resulting in more than 6,000 houses.
    - Trained the YOLO model once again on 20 epochs, which takes around 2 hours each time.
    - Iterated on the confidence threshold and rules for the model, trying to optimise error without overfitting.
    - Improved the architecture of the prototype codebase so that everything is more modular and the model can be swapped out in the future if needed.
    - Implemented the JSON schema that Nakshatra made so the model returns in the right format and the prototype graphics engine works.
- Daniel Hargrave:
    - Finished the first version of the YOLO model.
        - Used the CubiCasa5k dataset.
        - Keeps track of the walls, doors, windows, and stairs. Assigns a confidence level to everything.
        - Trained with 45 epochs which takes like 10 hours to train with the dataset. It didn’t plateau with this so the model can be improved with

additional epochs.

- The model managed to perform fairly well with the buildertrend images, which were not included as part of the training set for the model.
- Currently walls don't always fully connect, this is a simple problem to solve that will be described later.
- More stuff needs to be added for this to improve.
- Below is an example of the labels that it assigns.



- Jase Beaubien:
  - Coordinated high-level plans.
  - Helped theorize computer vision solutions.

Pending issues

- Nakshatra Gupta:
  - The schema was built from a single blueprint, so it likely needs changes to cover features in the other plans, such as multi-plane roofs, porches and stepped floor levels. It is also not yet confirmed that the YOLO-based computer vision model can output every field the schema requires, so the schema may need to shrink or

change to match what is detectable.

- Rishiraj Parmar: N/A
- Ryan McNab:
    - There are missing walls in areas where there are clearly walls and there are added walls in locations where furniture is messing with the line reading. More blueprints are needed to make the model more consistent.
- Daniel Hargrave:
    - Really need to add a post processing step that snaps the walls together.
    - Cubicasa5k dataset isn't allowed on commercial products. It will need to be replaced later on that is a part of the actual commercial version.
- Jase Beaubien:
    - Don't have models to work with.

Individual Contributions

NAME	**Individual Contributions** *(Quick list of contributions. This should be short.)*	**Hours this week**	**HOURS cumulative**
Nakshatra Gupta	Took ownership of the shared JSON schema, produced a corrected v0.2 package and an explanatory handout, and contributed to the scope discussion.	2	24
Rishiraj Parmar	Worked on floor elevation calculations for multi-story reconstruction.	2	25
Ryan McNab	Trained a few different models trying to optimize error loss between them.	5	33
Daniel Hargrave	Worked on the model significantly. Got an initial version that is fairly promising, but has several ways that it can be improved.	7	30
Jase Beaubien	Planning and research for graphics.	4	25

Comments and Extended Discussion

Plans for the upcoming week

- Nakshatra Gupta:
    - Collect feedback on the schema from the Computer Vision and Graphics teams. Revise it to cover features from the other three sample plans.
- Rishiraj Parmar:
    - Connect the floor elevation calculations to the shared JSON input so the prototype can read story heights and calculate floor positions without manually entering the values.
- Ryan McNab:
    - Start on the real project repository UI and keep the architecture very modular so each module can be imported into the project with minimal work.
- Daniel Hargrave:
    - Modify the model to handle angles more, since the model currently isn't able to really.
    - Retrain the model so that it uses larger images and more epochs.
    - Will likely add a few of the buildertrend images to the training process so the model is able to view a few of the projects that have larger images.
    - Add a post-processing step that "snaps" walls together that are close to one another.
    - Maybe implement a thing where no room can exist without an opening or door. This one will require significantly more time to be effective.
- Jase Beaubien:
    - Simulate basic models in the program.